Software Requirements Specification (SRS)

Vehicle Parking System

Project: Vehicle Parking System

Version: 1.1

Date: 13-11-2026 (v1.0: 06-09-2026)

Status: Baselined v1.1 (Sprint 3 traceability update)

Revision history

Version	Date	Author	Change summary
1.0	06-09-2026	Project Team	Initial SRS draft with diagrams
1.1	13-11-2026	Daruru Bhuvan Sai (QA Lead)	Detailed requirement tables with measurable acceptance criteria for 4.3-4.5 (VPS-F-020..023, 030..032, 040..043); new 4.6 Driver Account Management (VPS-F-050, 051) formalising the implemented registration story (Jira VPS-2); RTM (8) extended to every requirement with Sprint 3 test status. No existing requirement changed.

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Vehicle Parking System (VPS). It defines the functional and non-functional requirements, external interfaces, and verification criteria for a system that manages parking slot allocation, vehicle entry/exit, and vehicle tracking across one or more parking facilities.

1.2 Scope

The system covers driver-facing functions (registration, slot search, online reservation, payment, ticketing) and staff-facing functions (lot configuration, occupancy monitoring, attendant management, reporting) for a single or multi-facility parking operator. It excludes physical construction/civil work of parking structures and the internal implementation of third-party payment gateways beyond their published APIs.

1.3 Audience

Developers, QA Engineers, System Integrators, Parking Facility Operators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: VPS (Vehicle Parking System), ANPR (Automatic Number Plate Recognition), QR (Quick Response code), API (Application Programming Interface), UI (User Interface), RFID (Radio-Frequency Identification), JWT (JSON Web Token), TLS (Transport Layer Security), RTO (Recovery Time Objective), RPO (Recovery Point Objective).

2. Overall description

2.1 Product perspective

The Vehicle Parking System is a web-based platform (built on the MERN stack, with a Python/Flask or Django variant as an alternative backend) that coordinates parking slot inventory, reservations, entry/exit gate events, and payments. It interacts with drivers via a web/mobile-responsive interface, with facility hardware (gate barriers, ANPR cameras or RFID/QR scanners) via a hardware-abstraction API, and with external payment gateways for transaction processing.

2.2 Major product functions (detailed)

- Driver registration, authentication, and profile/vehicle management
- Real-time search and display of available parking slots by facility/zone
- Online slot reservation and pre-booking with time-window locking
- Vehicle entry logging (manual, QR, or ANPR-based) and automatic slot assignment
- Vehicle exit processing with duration- and rate-based fee calculation
- Online and at-gate payment processing with receipt/ticket generation
- Parking lot, zone, and slot configuration and pricing management (admin)
- Real-time occupancy monitoring dashboard and analytics/reporting
- Attendant account management and shift-based access control
- Manual gate override and exception handling for hardware failures

2.3 User roles and characteristics (expanded)

- Driver / Vehicle Owner: general public, expects a fast booking flow and clear fee breakdown; may have limited technical proficiency.
- Parking Attendant: on-site staff, needs a simple interface for manual entry/exit overrides and real-time occupancy views.
- Admin / Facility Operator: manages lot configuration, pricing rules, staff accounts, and reviews analytics/reports.

2.4 Operating environment

Cloud-hosted backend (Node.js/Express with MongoDB, or Python Flask/Django with a relational database) served over HTTPS; a responsive web front end (React for the MERN variant); on-premise gate controllers and sensors connected over a local network or the internet; mobile data/Wi-Fi connectivity for driver-facing apps.

2.5 Constraints

- Must integrate with at least one third-party payment gateway (e.g., Razorpay/Stripe sandbox) via published REST APIs.
- Must support concurrent slot booking without double-allocation (atomic slot-locking required).
- Must comply with applicable data-protection norms for storage of personal and vehicle data.
- Hardware interface layer must tolerate intermittent connectivity to gate controllers.

[Note: This SRS defines at least 15 functional requirements, 5 non-functional requirements, 2 security objectives, and 5 security requirements, as required for the assignment.]

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web interface (desktop and mobile browser) with a clear menu structure for search, booking, and payment. A simplified kiosk-style UI is provided at physical entry/exit points for attendants. Accessibility: legible high-contrast text and screen-reader-friendly labels on key actions.

3.2 Hardware interfaces

- Boom barrier / gate controller (relay or serial/network interface)
- ANPR camera or QR/RFID scanner at entry and exit lanes
- Slot-level occupancy sensors (ultrasonic/IR) — optional, for real-time slot status
- Receipt/ticket printer or digital ticket display
- Status indicator (LED/display) showing full/available lots

3.3 Software interfaces

- Payment Gateway API (HTTPS/JSON) for processing online and at-gate payments and refunds
- SMS/Email notification API for booking confirmations and receipts
- Mapping/geolocation API (optional) for showing nearby facilities
- Internal REST API between front end, backend, and the hardware-abstraction service

3.4 Communications

- TLS 1.2+ enforced for all client-server and server-to-gateway communication
- Retry/backoff strategy for gate-controller commands with local queuing during network loss
- Webhook-based asynchronous confirmation from the payment gateway, with signature verification

4.System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow VPS-F-###.

4.1 Vehicle Entry & Slot Allocation

Description: Log vehicle entry at the gate (manual, QR, or ANPR) and automatically allocate the nearest available slot matching the vehicle type.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-001	The system shall capture vehicle entry via ANPR, QR ticket, or manual attendant input and create a session record.	Functional	High	Facility Operations	AC-VPS-F-001: Entry event creates a session with timestamp and vehicle ID. Test: TC-Entry-01	Requires camera/scanner integration
VPS-F-002	The system shall automatically allocate the nearest available slot matching the vehicle's size/type and reserve it against the session.	Functional	High	Business	AC-VPS-F-002: Allocated slot status changes to Occupied and is not offered to another vehicle. Test: TC-Entry-02	Requires slot-locking logic
VPS-F-003	The system shall reject entry and display a 'Facility Full' message when no matching slot is available.	Functional	High	Operations	AC-VPS-F-003: Barrier remains closed and message is shown when occupancy = capacity. Test: TC-Entry-03	Depends on real-time occupancy count
VPS-F-004	The system shall print or issue a digital entry ticket (QR code) containing session ID, entry time, and slot number.	Functional	Medium	Customer Experience	AC-VPS-F-004: Ticket contains all required fields and scans correctly at exit. Test: TC-Entry-04	Requires printer/QR generation

4.2 Vehicle Exit & Fee Calculation

Description: Process vehicle exit, calculate the parking fee based on duration and rate rules, and confirm payment before releasing the barrier.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-010	The system shall calculate the parking fee based on duration parked and the applicable rate	Functional	High	Business	AC-VPS-F-010: Fee matches duration x rate within rounding rules. Test: TC-Exit-01	Depends on pricing configuration

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	card for the vehicle type and zone.					
VPS-F-011	The system shall verify payment completion (online pre-payment or at-gate payment) before releasing the exit barrier.	Functional	High	Security/Business	AC-VPS-F-011: Barrier opens only after payment status = Confirmed. Test: TC-Exit-02	Requires Payment Gateway API
VPS-F-012	The system shall release the allocated slot back to the available pool immediately upon confirmed exit.	Functional	High	Operations	AC-VPS-F-012: Slot status changes to Available within 2 seconds of exit confirmation. Test: TC-Exit-03	Updates real-time occupancy dashboard
VPS-F-013	The system shall generate a digital/printed receipt showing entry time, exit time, duration, and amount charged.	Functional	Medium	Customer Experience	AC-VPS-F-013: Receipt fields match session record. Test: TC-Exit-04	Requires printer/email service

4.3 Slot Reservation / Online Pre-booking

Requirements: Allow registered drivers to search available slots by facility/zone/date-time and reserve a slot in advance, with the reservation held for a configurable grace period. Reject bookings that exceed facility capacity or fall outside operating hours. IDs: VPS-F-020..VPS-F-023.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-020	The system shall let a registered driver reserve a slot at a chosen facility (optionally a zone) for a time window, locking the slot atomically so it cannot be double-booked.	Functional	High	Drivers	AC-VPS-F-020: Response has reservation ID, slot, status HELD and hold expiry; the slot is not offered to others for that window; of concurrent requests for the last slot exactly one succeeds. Test: TC-Res-01	Atomic slot locking (2.5)
VPS-F-021	The system shall show, in real time, the slots that are free for the whole requested window, filtered by facility, zone and vehicle type and ordered nearest first.	Functional	High	Drivers	AC-VPS-F-021: Only free matching slots are listed; back-to-back windows are allowed; windows in the past or outside opening hours return 422. Test: TC-Res-02	Live updates via server-sent events
VPS-F-022	The system shall hold a reservation until a configurable grace period after its start time and release the slot if the vehicle has not entered.	Functional	Medium	Operations	AC-VPS-F-022: After the grace period (default 15 min, per facility) the reservation becomes EXPIRED and the slot	Grace period configured per facility (4.5)

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
					is available again. Test: TC-Res-03	
VPS-F-023	The system shall reject bookings that exceed facility capacity or fall outside operating hours.	Functional	High	Business	AC-VPS-F-023: 409 when no matching slot is free for the window; 422 outside opening hours; capacity is never oversold. Test: TC-Res-04	Depends on VPS-F-040 opening hours

4.4 Payment Processing

Requirements: Support online payment for reservations and at-gate payment for walk-in exits via an integrated payment gateway; support refunds for cancelled reservations within policy; log all transactions with gateway reference IDs. IDs: VPS-F-030..VPS-F-032.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-030	The system shall accept online payment for reservations and at-gate payment for exits through the payment gateway, and treat a payment as complete only after the gateway's signed confirmation.	Functional	High	Business / Security	AC-VPS-F-030: Each payment stores the gateway reference; only CONFIRMED payments settle a booking or session; a webhook with a bad signature gets 401 and changes nothing. Test: TC-Pay-01	Payment gateway API (3.3); VPS-SR-004
VPS-F-031	The system shall refund in full a paid reservation cancelled before the facility's refund cutoff, and cancel without refund after it.	Functional	Medium	Customer Experience	AC-VPS-F-031: Cancelled before the cutoff (default 60 min before start): refund recorded, status REFUNDED; after it: status CANCELLED, no refund. Test: TC-Pay-02	Cutoff configured per facility
VPS-F-032	The system shall log every payment and refund with amount, status, gateway reference and time, and let admins list them by type and date.	Functional	Medium	Finance / Audit	AC-VPS-F-032: The admin transaction list contains every payment and refund with numeric amount, status, gateway reference and creation time. Test: TC-Pay-03	Retained per VPS-NF-004

4.5 Admin, Attendant & Reporting

Requirements: Provide an admin console for configuring parking lots, zones, slots, and pricing rules; provide attendant accounts with manual entry/exit override capability; provide occupancy, revenue, and usage reports exportable in common formats. IDs: VPS-F-040..VPS-F-043.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-040	The system shall provide an admin console to configure facilities (opening hours, grace and refund settings), zones, gates and slots, including deactivating slots.	Functional	High	Facility Operator	AC-VPS-F-040: Changes apply to the next allocation; an inactive slot is never allocated; deactivating a slot that is occupied or booked is refused (409). Test: TC-Admin-01	Admin role only (VPS-SR-003)
VPS-F-041	The system shall let an admin set hourly rates per facility, zone and vehicle type.	Functional	High	Business	AC-VPS-F-041: A new rate applies to vehicles entering after the change; vehicles already parked keep the rate from their entry time. Test: TC-Admin-02	Used by VPS-F-010
VPS-F-042	The system shall provide attendant accounts, created by an admin, that can monitor occupancy and open or close a gate manually when hardware fails; every override is audited.	Functional	High	Operations	AC-VPS-F-042: Created accounts have role ATTENDANT; an OPEN/CLOSE override is delivered to the gate (or queued while it is offline) and audited with user, gate, action and reason. Test: TC-Admin-03	Hardware abstraction (3.2); network-loss queuing (3.4)
VPS-F-043	The system shall produce occupancy, revenue and usage reports for a date range, exportable as CSV.	Functional	Medium	Facility Operator	AC-VPS-F-043: Report totals match the recorded transactions; CSV export returns text/csv with a row per day and a TOTAL row. Test: TC-Admin-04	Range up to 366 days

4.6 Driver Account Management

Requirements: Let drivers register themselves with their vehicles and manage their own profile. Added in v1.1 to give the implemented registration story (Jira VPS-2) its own IDs. IDs: VPS-F-050..VPS-F-051.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-050	The system shall let a driver self-register with name, email, mobile number, password and	Functional	High	Drivers	AC-VPS-F-050: 201 with a login token; duplicate email (any	Staff accounts are created by

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	optional vehicles; self-registration always creates a Driver account.				case) or number plate gets 409 and creates nothing; invalid fields get 400 with per-field details; password needs at least 8 characters with a letter and a number. Test: auth.api.test.js; UI walkthrough	an admin only (VPS-F-042)
VPS-F-051	The system shall let a logged-in driver view and edit their profile, change their password and add or remove their vehicles.	Functional	Medium	Drivers	AC-VPS-F-051: Name and phone can change but email and role cannot; password changes need the current password; a driver cannot remove another driver's vehicle. Test: auth.api.test.js	Requires VPS-F-050

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs VPS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
VPS-NF-001	End-to-end entry/exit transaction (barrier command to confirmation) shall complete within 3 seconds for 90% of transactions under normal load.	Performance	High	90th percentile <= 3s in load test. Test: TC-Perf-01
VPS-NF-002	The system shall provide 99.5% monthly availability for the reservation and payment services; scheduled maintenance windows excluded.	Reliability	High	Uptime reports show >=99.5% per month. Test: Ops reports
VPS-NF-003	All payment and personal data must be transmitted and stored in compliance with applicable data-protection requirements; card data must never be stored in plaintext.	Security/Compliance	High	Security review checklist pass. Test: TC-Sec-01
VPS-NF-004	The system shall retain transaction and access logs for at least 1 year for audit and dispute-resolution purposes.	Audit/Data Retention	Medium	Automated archival and retrieval verified. Test: TC-OPS-01
VPS-NF-005	The system shall support horizontal scaling to accommodate at least 10x the baseline concurrent users without code changes.	Scalability/Maintainability	Medium	Load test at 10x baseline passes without errors. Test: TC-Perf-02

5.1 Security

5.1.1 Security Objectives

- Protect driver personal data, vehicle information, and payment details from unauthorized access or disclosure.
- Ensure the integrity of parking transaction records (tickets, sessions, payments) to prevent tampering or fraudulent fee manipulation.
- Ensure availability of the entry/exit gate control and reservation services against disruption, including graceful degradation during network loss.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
VPS-SR-001	TLS 1.2+ shall be mandatory for all network connections between client, server, and payment gateway.	Security	High	TLS scan confirms no connections below 1.2. Test: TC-Sec-02
VPS-SR-002	The system shall hash and salt all stored user passwords using a industry-standard algorithm (e.g., bcrypt/argon2).	Security	High	Password DB review confirms no plaintext passwords. Test: TC-Sec-03
VPS-SR-003	The system shall enforce role-based access control (RBAC) so attendants and admins can only access functions permitted for their role.	Security	High	Access-control test matrix passes for all roles. Test: TC-Sec-04

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
VPS-SR-004	The system shall never store full payment card numbers; only tokenized references from the payment gateway shall be persisted.	Security	High	Database review confirms only tokens present. Test: TC-Sec-05
VPS-SR-005	The system shall log and alert on repeated failed login attempts and lock the account after 5 consecutive failures.	Security	Medium	Account locks after 5 failed attempts; alert generated. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

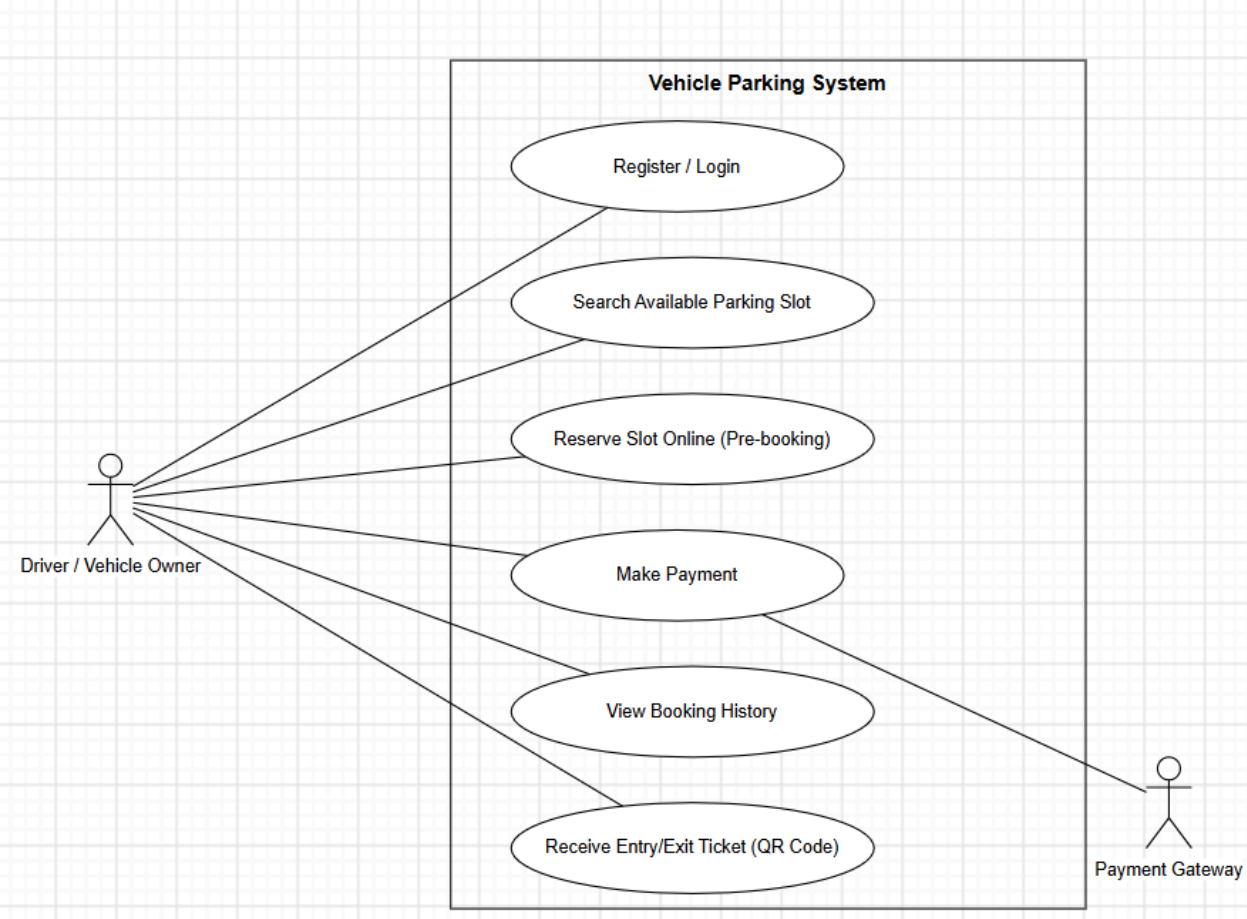
- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed.

- Acceptance test suites: Registration/Authentication, Entry & Slot Allocation, Exit & Fee Calculation, Reservation, Payment, Reporting, Performance, Security, and Accessibility tests.

7. System models and diagrams

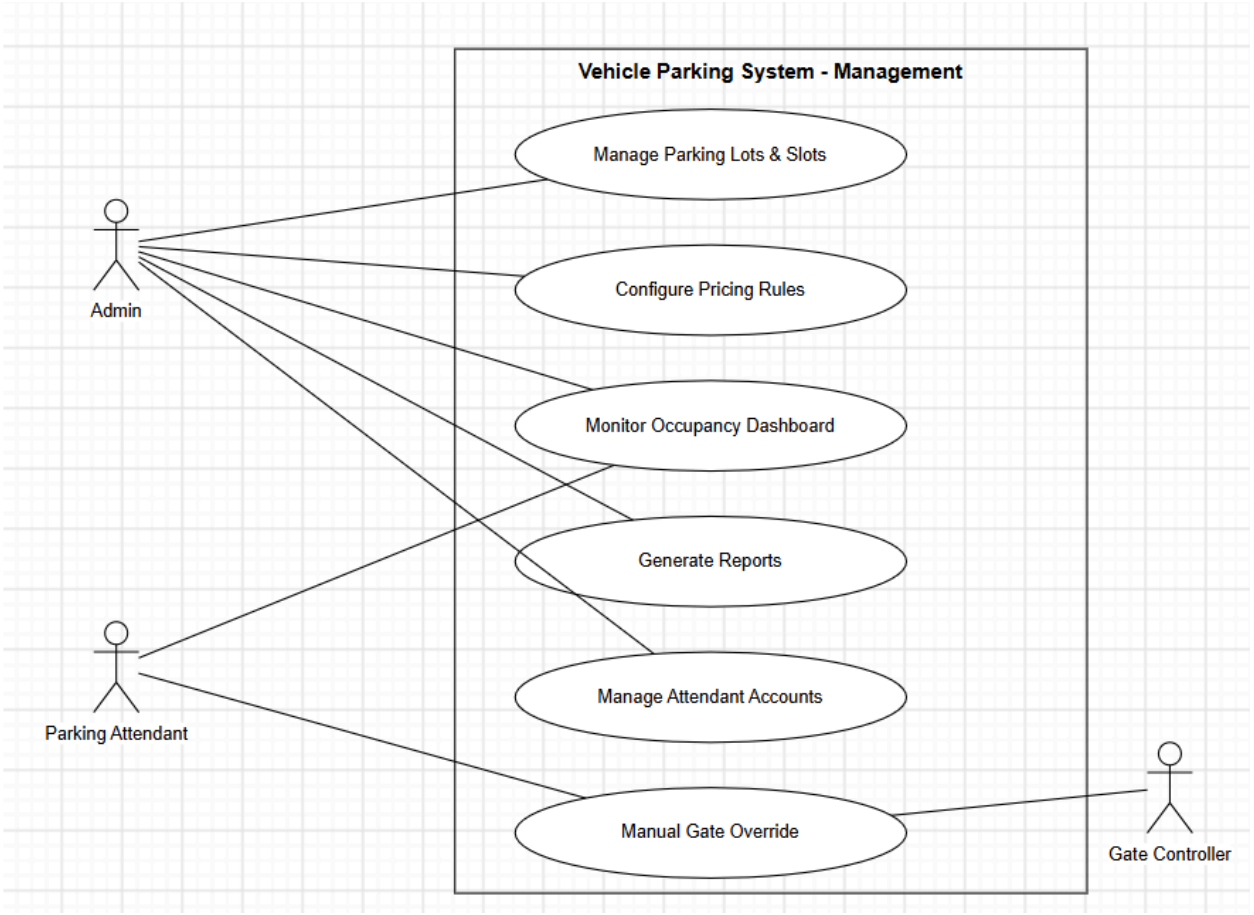
7.1 UML Use-Case Diagram 1 — Driver / Vehicle Owner

This diagram shows the primary interactions of a driver with the Vehicle Parking System, from registration through slot search, reservation, payment, and ticket issuance. The Payment Gateway is modelled as an external actor.



7.2 UML Use-Case Diagram 2 — Admin / Parking Attendant

This diagram shows the management-side interactions: the Admin configures lots, pricing, and staff accounts and reviews reports; the Parking Attendant monitors occupancy and can manually override the gate; the Gate Controller is modelled as an external actor triggered by the manual override use case.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
VPS-F-001	Capture vehicle entry	4.1 / DS-Entry-01	EntryModule	TC-Entry-01	P	Acceptance + UI 13-11-2026
VPS-F-002	Auto slot allocation	4.1 / DS-Entry-02	AllocationModule	TC-Entry-02	P	Incl. concurrency race
VPS-F-010	Calculate parking fee	4.2 / DS-Exit-01	BillingModule	TC-Exit-01	P	Started-hour rounding
VPS-F-011	Verify payment before exit	4.2 / DS-Exit-02	PaymentModule	TC-Exit-02	P	402 until signed confirmation
VPS-F-020	Online slot reservation	4.3 / DS-Res-01	ReservationModule	TC-Res-01	P	UI booking verified
VPS-NF-001	Transaction response time	5 / DS-Perf-01	WebUI / API	TC-Perf-01	P	Local p90 79-91 ms; deployed JMeter pending
VPS-SR-001	TLS 1.2+ enforcement	5.1.2 / DS-Sec-01	All Modules	TC-Sec-02	A	Local TLS checks; deployed scan pending
VPS-F-003	Facility Full rejection	4.1 / DS-Entry-02	EntryModule	TC-Entry-03	P	
VPS-F-004	Entry ticket (QR)	4.1 / DS-Entry-01	TicketModule	TC-Entry-04	P	
VPS-F-012	Slot release on exit	4.2 / DS-Exit-01	AllocationModule	TC-Exit-03	P	Released in the exit request
VPS-F-013	Exit receipt	4.2 / DS-Exit-01	ReceiptModule	TC-Exit-04	P	
VPS-F-021	Real-time slot search	4.3 / DS-Res-01	ReservationModule	TC-Res-02	P	
VPS-F-022	Grace-period hold	4.3 / DS-Res-01	ReservationModule	TC-Res-03	P	
VPS-F-023	Capacity / hours rules	4.3 / DS-Res-01	ReservationModule	TC-Res-04	P	
VPS-F-030	Online and at-gate payment	4.4 / DS-Exit-02	PaymentModule	TC-Pay-01	P	DEF-001: admin denied at-gate payment (VPS-47)
VPS-F-031	Cancellation refund	4.4 / DS-Exit-02	PaymentModule	TC-Pay-02	P	
VPS-F-032	Transaction log	4.4 / DS-Exit-02	PaymentModule	TC-Pay-03	P	
VPS-F-040	Lot / zone / slot config	4.5	AdminModule	TC-Admin-01	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
VPS-F-041	Pricing rules	4.5	AdminModule	TC-Admin-02	P	
VPS-F-042	Attendants and gate override	4.5	AdminModule / Hardware	TC-Admin-03	P	
VPS-F-043	Reports with CSV export	4.5	ReportingModule	TC-Admin-04	P	
VPS-F-050	Driver self-registration	4.6	AuthModule	auth.api.test.js	P	Added v1.1
VPS-F-051	Profile and vehicles	4.6	AuthModule	auth.api.test.js	P	Added v1.1
VPS-NF-002	99.5% monthly availability	5	Ops	TC-OPS-02	N	Needs a month of monitoring
VPS-NF-003	Data protection	5 / 5.1	All Modules	TC-Sec-01	P	
VPS-NF-004	Logs kept 1 year	5	AuditModule	TC-OPS-01	P	Synthetic timestamps
VPS-NF-005	10x horizontal scale	5	API / DB	TC-Perf-02	A	10x on one instance passes; multi-instanc e not run
VPS-SR-002	Salted password hashes	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	P	bcrypt
VPS-SR-003	Role-based access control	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-04	P	
VPS-SR-004	Tokenized card data only	5.1.2 / DS-Sec-01	PaymentModule	TC-Sec-05	P	
VPS-SR-005	Lockout and alert	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-06	P	Plus per-IP throttle

Status legend (v1.1, 13-11-2026): N = not run, P = passed, A = accepted with documented limits (partially verified). Evidence: Test_Closure_Report.md and docs/evidence/closure-2026-11-13 in the SE_mini_project repository; deviations from this SRS are listed in docs/Design_Deviations.md.